

1.3 Chemical Bonding

Question Paper

Course	CIEA Level Chemistry
Section	1. Physical Chemistry
Topic	1.3 Chemical Bonding
Difficulty	Medium

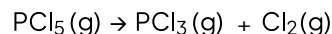
Time allowed: 20

Score: /9

Percentage: /100

Question 1

The following equation shows the dissociation equilibrium of PCl_5 .



The percentage yield of PCl_3 differs with temperature.

At 160°C PCl_3 yield is 13% and at 300°C yield is 100%.

	The reaction is	Shape of PCl_3 molecule
A	Exothermic	Pyramidal
B	Exothermic	Trigonal planar
C	Endothermic	Pyramidal
D	Endothermic	Trigonal planar

[1 mark]

Question 2

N_2 is a covalently bonded molecule that makes up 78% of gas in the air we breathe.

Which of the following is not present in an N_2 molecule?

- A. Two lone pairs of electrons
- B. Three shared pairs of electrons
- C. Three π bonds
- D. 180° bond angle

[1 mark]

Question 3

Which of the following elements has electronegativity most similar to beryllium?

- A. Boron
- B. Aluminium
- C. Silicon
- D. Indium

[1 mark]

Question 4

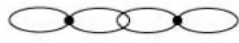
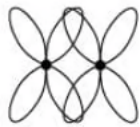
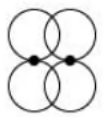
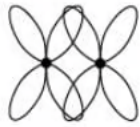
Three lone pairs of electrons can be found around the central atom of which of the following ions?

- A. PF_6^-
- B. ClF_2^-
- C. BF_2^-
- D. SF_5^-

[1 mark]

Question 5

Which diagrams correctly illustrate the formation of a σ bond and a π bond from the overlap of electron orbitals?

	σ	π
A		
B		
C		
D		

[1 mark]

Question 6

Which of the following changes of state shows only dispersion Van der Waals' forces being overcome?

- A. $\text{C}_3\text{H}_6(\text{l}) \rightarrow \text{C}_3\text{H}_6(\text{s})$
- B. $\text{CO}_2(\text{l}) \rightarrow \text{CO}_2(\text{g})$
- C. $\text{H}_2\text{O}(\text{s}) \rightarrow \text{H}_2\text{O}(\text{l})$
- D. $\text{C}_2\text{H}_5\text{OH}(\text{l}) \rightarrow \text{C}_2\text{H}_5\text{OH}(\text{g})$

[1 mark]**Question 7**

Before the 1820s scientists believed that inorganic and organic molecules were fundamentally different and could not be interchanged, the experiment by Francis Wöhler changed this theory as he was able to produce organic urea, $\text{CO}(\text{NH}_2)_2$ from inorganic ammonium cyanate.

What is the formula of the cyanate ion in ammonium cyanate?

- A. CO^-
- B. NO^-
- C. CNO^-
- D. CNO^{2-}

[1 mark]**Question 8**

"The bondX.... between the hydrogen and halide in HBr isY.... than in HI."

Which pair of words does not correctly complete the sentence above.

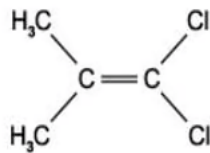
	X	Y
A	Length	Shorter
B	Energy	Greater
C	Angle	Smaller
D	Polarity	Greater

[1 mark]

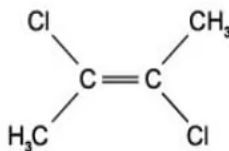
Question 9

Which of the following molecules has the biggest overall dipole?

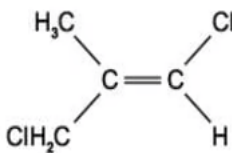
A.



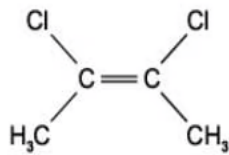
B.



C.



D.



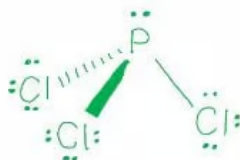
[1 mark]

Model Answers: Medium

1

The correct answer is **C** because:

- If the reaction has a higher yield at a higher temperature, then the reaction will be endothermic.
- The **VESPR theory** states that electron pairs **repel** each other whether or not they are in bond pairs or lone pairs; therefore the electrons will spread as much as possible to **reduce repulsion**.
 - This creates the shape of the molecules based on the atoms and lone pairs present.
- PCl_3 has three atoms around the central atom with one lone pair on the phosphorus making it trigonal pyramidal in shape with a bond angle of 107.5° .



A & B are incorrect as if the reaction was exothermic it would produce less yield at higher temperatures.

D is incorrect as the phosphorus atom has a lone pair of electrons, making the bond angle 107° not 109.5° .

2

The correct answer is **C** because:

- Nitrogen has 5 valence electrons; three will take part in covalent bonds, leaving one lone pair of electrons on each atom.
- Between nitrogen atoms, there is a triple bond (3 covalent bonds, each formed

from an electron from each nitrogen atom).

- Sigma (σ) and pi (π) bonds are formed by the overlap of atomic orbitals.
- Sigma bonds are formed by end to end orbital overlapping.
- Pi bonds are when the lobe of one atomic orbital overlaps with another.
- A triple bond has one sigma and two pi bonds.

A is incorrect as each atom of nitrogen has a lone pair of electrons.

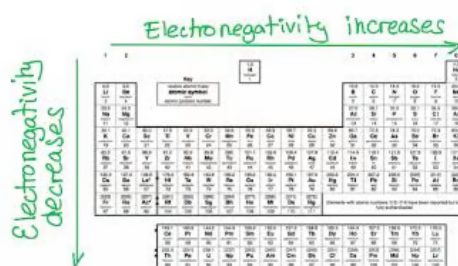
B is incorrect as there are three covalently bonded electrons in N_2 .

D is incorrect as the bond angle in N_2 will be 180° as this is a linear molecule.

3

The correct answer is **B** because:

- The electronegativity of an atom is affected by the atomic number and the distance between the valence electrons and the nucleus.
 - Atoms with high electronegativity attract electrons strongly.
 - Atoms with low electronegativity attract electrons weakly.
- The Pauling electronegativity scale is based on the bond-energy calculations for different electron configurations.
- The electronegativity increases as you move to the right of the periodic table.
- The electronegativity decreases as you move down the groups.
- The electronegativity of beryllium is 1.5; beryllium is top left of the periodic table.
- The electronegativity of aluminium is 1.6, aluminium is a larger atom and further to the right.



A is incorrect as the electronegativity of boron is 2.0, boron is a smaller atom than aluminium and further to the right.

C is incorrect as the electronegativity of silicon is 1.8, silicon is a larger atom and further to the right than beryllium.

D is incorrect as the electronegativity of indium is 1.7; indium is a larger atom and closer to the right than beryllium.

4

The correct answer is **B** because:

- **Lewis dot** structures show the bonding pairs and lone pairs in a structure, to draw a Lewis dot structure follow these steps:
 - Work out the total number of valence electrons, for cations subtract one for anions add one for each charge
 - Draw a skeleton structure of the molecules around the central atom – the least electronegative atom should be in the middle, connect with a single bond and subtract the bonding electrons from the total
 - Draw in the lone pairs to complete the octet around each terminal atom
 - Place the remaining electrons on the central atom
 - Re-arrange the electrons of the outer shells to make multiple bonds with the central atoms to obtain octets where possible
- Total electrons = 22
- Each bond uses two electrons; $22 - 4 = 18$
- Fill the octet for each fluorine; $18 - 12 = 6$
- Chlorine is in period 3 so can have an expanded octet, so the remaining electrons

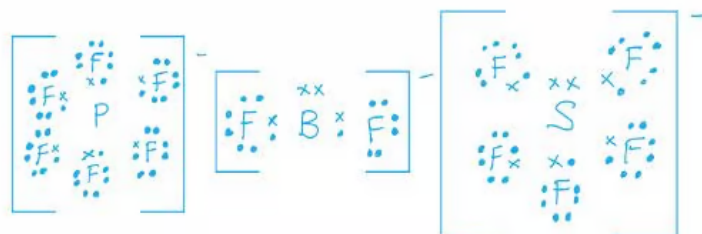
are placed on chlorine



- Each bond uses two electrons; $22 - 4 = 18$
- Fill the octet for each fluorine; $18 - 12 = 6$
- Chlorine is in period 3 so can have an expanded octet, so the remaining electrons are placed on chlorine



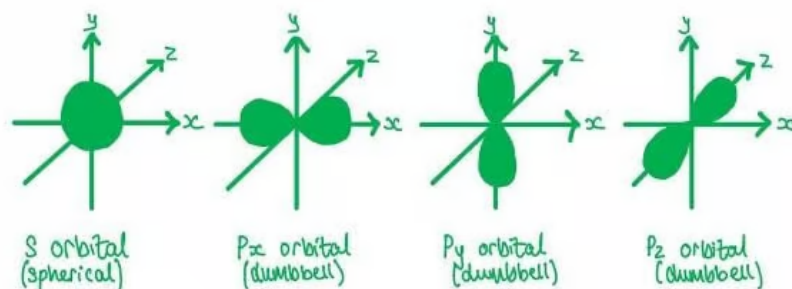
A, C & D are incorrect as none of the other structures has three lone pairs around the central atom.



5

The correct answer is **B** because:

- A sigma (σ) bond is formed by the head-on overlap of; two s orbitals, an s and a p orbital or two p orbitals.
- A pi (π) bond is formed by the sideways overlap of 2 p orbitals.
- S orbitals are spherical in shape, and p orbitals are dumbbell in shape.



- Answer **B** shows two p orbitals meeting head-on to create a σ bond and two p orbitals meeting sideways to show a π bond.

A is incorrect as the pi bond is not formed with an s and p orbital.

C & D are incorrect as the bond in the π bond column is a delta bond. Where 4 lobes on one atomic orbital overlap four lobes of the other atomic orbital.

6

The correct answer is **B** because:

- Van der Waals interactions are the weakest of all intermolecular attractions between molecules.
- There are three types of Van der Waals forces; dispersion, dipole-dipole, and hydrogen bonding.
- Dispersion (or London forces) are temporary or induced dipoles in molecules that attract each other weakly, the molecular size and shape will affect the strength of these forces and the boiling points.
- Some molecules can have permanent dipoles and create dipole-dipole interactions, molecules with permanent dipoles have boiling points higher than ones with dispersion forces alone.
- The molecular size of carbon dioxide means that it will have weak dispersion forces and therefore a very low boiling point.

A is incorrect as dispersion forces are being formed as propane changes from a liquid to a solid.

C is incorrect as water has hydrogen bonds between the molecules which are being broken in this example.

D is incorrect as ethanol has permanent hydrogen bonds between the molecules which are being broken when ethanol vaporises.

7

The correct answer is **C** because:

- This reaction is known as Wöhler synthesis, creating urea from ammonium cyanate.
- Urea contains; one carbon atom, one oxygen atom, two nitrogen atoms, and four hydrogen atoms.
- The ammonium ion NH_4^+ contains, one nitrogen atom, and four hydrogen atoms.
- So, therefore, the cyanate ion must be what is left; one carbon atom, one oxygen atom and one nitrogen atom.
- The ammonium ion has a +1 charge so the cyanate ion must have a -1 charge.

A is incorrect as this is carbon monoxide and does not have the nitrogen needed.

B is incorrect as this is a nitric oxide ion and doesn't contain the carbon needed.

D is incorrect as this has the incorrect charge of 2-.

8

The correct answer is **C** because the molecules have only two atoms, so there is no bond angle, therefore this incorrectly completes the sentence.

A is incorrect as length of the bond in HBr is shorter than in HI due to the shell number increasing down the halide group.

B is incorrect as the energy needed to break the bond decreases as the bond length increases.

D is incorrect as bromine is more electronegative than iodine creating a higher polarity in the HBr molecule.

9

The correct answer is **A** because:

- All of the molecules shown in the question are isomers of $\text{C}_4\text{H}_6\text{Cl}_2$.
- Chlorine is the most electronegative atom in the molecule.
- The electronegativity increases as you move to the right of the periodic table.

- The electronegativity decreases as you move down the groups.
- The electronegativity of chlorine is 3.16.

